

LAGOON CITY, ON, Canada

WMO: 712820

Lat: 44.548N

Lon: 79.221W

Elev: 724

StdP: 14.32

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-5.1	-17.7	2.1	-9.9	-11.7	3.0	-4.2	33.5	18.4	30.0	19.4	6.6	20	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	81.3	72.6	79.0	71.1	77.1	69.9	75.3	78.9	73.4	77.0	71.6	75.6	8.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.2	131.0	77.5	72.1	121.9	75.9	70.1	114.0	74.4	39.3	79.0	37.5	77.0	35.8	75.7	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.5	25.0	22.0	DB	-19.6	87.4	6.0	3.8	-23.8	90.1	-27.3	92.3	-30.7	94.4	-35.0	97.2	(4)
(5)			WB	-19.8	77.7	5.9	3.1	-24.1	79.9	-27.5	81.6	-30.8	83.4	-35.0	85.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.4	16.8	18.7	27.5	40.5	54.3	63.9	68.9	67.7	60.7	48.9	37.0	26.1	(6)	
	DBStd	20.15	12.61	10.91	10.48	7.66	7.39	6.20	5.20	5.01	6.81	7.67	8.39	10.01	(7)	
	HDD50	4190	1028	876	698	294	40	0	0	0	4	115	394	741	(8)	
	HDD65	7877	1493	1296	1162	734	341	91	21	31	162	499	841	1206	(9)	
	CDD50	2149	0	0	1	10	173	419	587	548	326	82	3	0	(10)	
	CDD65	359	0	0	0	0	9	60	143	113	33	1	0	0	(11)	
	CDH74	1329	0	0	0	1	26	217	605	389	90	1	0	0	(12)	
	CDH80	139	0	0	0	0	2	22	74	35	6	0	0	0	(13)	
(14) Wind	WSAvg	10.1	10.9	10.3	9.9	10.2	9.1	8.6	8.9	8.8	9.1	11.2	12.1	12.0	(14)	
(15) Precipitation	PrecAvg	37.8	3.5	2.4	2.3	2.7	3.7	3.0	3.5	2.8	3.3	3.3	3.6	3.6	(15)	
	PrecMax	44.1	6.7	4.2	4.7	5.4	6.1	4.8	5.3	4.5	5.1	5.6	7.2	5.5	(16)	
	PrecMin	32.6	1.7	0.9	0.7	0.9	1.7	1.2	0.9	0.9	1.1	0.8	1.9	1.7	(17)	
	PrecStd	2.9	1.3	0.9	1.2	1.1	1.1	1.0	1.2	0.9	1.1	1.5	1.5	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.5	43.6	55.0	65.6	77.3	83.1	85.4	84.0	80.5	70.9	57.0	48.4	(19)	
		MCWB	42.9	41.3	48.9	53.9	67.9	72.6	76.4	74.5	70.9	65.3	53.0	46.0	(20)	
	2%	DB	38.8	39.5	47.7	58.5	72.7	79.2	81.9	80.1	76.4	66.6	53.9	44.5	(21)	
		MCWB	37.7	37.5	42.6	49.9	65.1	71.0	73.2	72.2	69.7	62.0	51.0	43.1	(22)	
	5%	DB	36.1	36.5	43.5	54.8	69.0	76.4	79.3	77.9	73.8	63.7	51.2	41.2	(23)	
		MCWB	35.0	34.8	40.2	48.0	61.8	69.1	71.5	70.6	68.2	59.7	48.4	39.6	(24)	
	10%	DB	33.7	34.0	40.4	51.3	65.3	73.5	77.3	75.9	71.4	60.3	48.8	38.3	(25)	
		MCWB	32.7	32.7	37.7	45.7	58.9	67.0	70.2	69.3	66.1	56.5	46.5	36.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.0	42.1	51.2	56.7	70.9	75.6	79.0	77.6	73.6	67.2	54.6	47.1	(27)	
		MCDB	43.4	43.1	53.1	63.1	74.1	80.3	83.3	80.5	76.9	69.8	56.2	48.2	(28)	
	2%	WB	38.1	37.8	44.0	52.4	67.1	72.8	76.0	74.6	71.0	63.1	51.6	43.2	(29)	
		MCDB	38.6	38.9	46.3	55.9	71.8	77.1	79.5	77.8	75.1	65.7	53.4	44.3	(30)	
	5%	WB	35.1	35.1	40.8	49.2	63.2	70.7	73.8	72.7	69.2	60.1	49.0	39.8	(31)	
		MCDB	35.8	36.4	42.8	53.1	67.2	74.8	77.4	76.2	73.0	63.1	50.7	41.1	(32)	
	10%	WB	32.8	32.8	38.0	46.4	60.1	68.6	71.8	70.8	67.1	57.2	46.7	36.9	(33)	
		MCDB	33.7	33.9	40.0	50.5	64.0	72.5	75.8	74.6	70.5	59.7	48.6	38.2	(34)	
(35) Mean Daily Temperature Range	MDBR		14.7	15.9	15.2	13.8	14.3	13.8	13.3	13.3	14.4	11.9	10.7	10.8	(35)	
	5% DB	MCDBR	11.6	14.9	16.8	19.6	18.6	16.9	14.9	15.0	16.3	15.3	13.7	10.3	(36)	
		MCWBR	11.3	13.6	12.9	12.3	10.9	10.1	8.8	8.9	10.3	11.5	12.0	9.8	(37)	
	5% WB	MCDBR	11.7	14.7	15.9	17.0	16.1	14.7	12.7	12.5	13.7	14.0	12.8	10.5	(38)	
		MCWBR	11.7	13.8	13.3	13.0	10.9	9.8	8.9	9.0	10.1	11.6	12.0	10.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.295	0.302	0.322	0.363	0.393	0.416	0.418	0.407	0.373	0.350	0.313	0.288	(40)	
	taud		2.356	2.329	2.398	2.348	2.310	2.267	2.282	2.300	2.390	2.441	2.496	2.439	(41)	
	Ebn at Noon		260	281	290	285	279	271	269	268	269	260	254	253	(42)	
	Edh at Noon		24	30	32	37	40	42	41	39	33	27	22	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		440	716	1136	1419	1703	1849	1879	1623	1284	747	450	326	(44)	
	RadStd		50	84	116	143	177	144	128	84	122	78	60	38	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page